

Lecture 3

Microeconomic Theory 3 – ECON 709
Fall 2026

based on: Rubinstein, lecture 2, lecture 4 (proof of Debreu's Theorem); MWG,
sections 1.B, 3.C

Recap' Lecture 2

- ▶ A function $u : X \rightarrow \mathbb{R}$ is a utility function representing preference relation $\succsim$ if, for all $x, y \in X$,
 $x \succsim y \Leftrightarrow u(x) \geq u(y)$.
- ▶ If a preference relation $\succsim$ has a utility representation, then there exist infinitely many utility functions which represent $\succsim$.
- ▶ Only if a preference relation $\succsim$ is rational, $\succsim$ can have a utility representation. Rationality is necessary for the existence of a utility representation!
- ▶ Under which conditions does a rational $\succsim$ have a utility representation?

If X is finite, then a rational $\succsim$ has a utility representation.

- ▶ a finite set is a set which has a finite number of elements; informally, one can count the elements and finish counting
- ▶ Intuition for why the above claim is true? Assigning utility values (from the real line with infinitely many numbers) in an order-preserving way to a finite set of ordered alternatives of course works!
- ▶ we present a very detailed proof of this claim, mostly to get into the habit of analytical precision
- ▶ the proof starts with a lemma regarding the existence of minimal elements
(an element $a \in X$ is *minimal* if $x \succsim a$ for any $x \in X$).

Lemma 1

If $\succsim$ is rational, then, in any finite set $A \subseteq X$, there is a minimal element (similarly, there is a maximal element).

Proof.

- ▶ By induction on the size of A .
- ▶ If A is a singleton (cardinality of A equal to 1), then, by reflexivity of $\succsim$, A 's only element is minimal.
- ▶ For the inductive step, let A be of cardinality $n + 1$ (where $n \geq 1$) and take any $x \in A$.
 - ▶ The set $A - \{x\}$ is of cardinality n and by the inductive assumption has a minimal element which we denote by y .
 - ▶ By completeness of $\succsim$, one of the two following will be satisfied:
 - ▶ If $x \succsim y$, then y is minimal in A .
 - ▶ If $y \succsim x$, then by transitivity $z \succsim x$ (because $z \succsim y$) for all $z \in A - \{x\}$, and thus x is minimal.



Proposition 4

If $\succsim$ is a rational preference relation on a finite set X , then $\succsim$ has a utility representation with values that are natural numbers.

(mapping a finite set in an order-preserving way onto a countably infinite set!)

Proof:

1. construct a sequence of "indifference" sets inductively:

- ▶ Let X_1 be the set of elements that are minimal in X .
- ▶ Assume we have constructed the non-empty sets $X_1, \dots, X_k$.
- ▶ If $X = X_1 \cup X_2 \cup \dots \cup X_k$, we are done.
- ▶ If not, define X_{k+1} to be the set of minimal elements in $X - X_1 - X_2 - \dots - X_k$. By the lemma, $X_{k+1} \neq \emptyset$. Because X is finite, we must be done after at most $|X|$ steps.
- ▶ Note that set X_k contains all elements which are at rank k in the preference relation (with 1 being the rank of least preferred alternatives in X).

2. define a utility function: let $u(x) = k$ if $x \in X_k$.
3. verify that u represents $\succsim$: for any $a, b \in X$,
 $a \succ b \Leftrightarrow a \notin X_1 \cup X_2 \cup \dots \cup X_{u(b)} \Leftrightarrow u(a) > u(b)$, and
 $a \sim b \Leftrightarrow a, b \in X_k \Leftrightarrow u(a) = u(b) = k$.



Existence of utility representations extends to all countable sets X

Every rational preference relation over a countable set X has a utility representation.

A set is countable if all its members can be enumerated, $\{x_n\}_{n=1,2,\dots}$, that is, we can count all its members with the positive integers.

A countable set can be finite or infinite (“you cannot finish counting”).

Proposition 5

If X is countable, then any rational preference relation $\succsim$ has a utility representation with a range $(-1, 1)$.

(mapping a countable set to an uncountable set (higher cardinality) in an order-preserving way!)

Proof of Proposition 5

- ▶ Let $\{x_n\}_{n=1,2,\dots}$ be an enumeration of all elements in X .
- ▶ Construct a utility function inductively.
 - ▶ Set $u(x_1) = 0$.
 - ▶ Assume that you have completed the definition of the values $u(x_1), \dots, u(x_{n-1})$ (all values in $(-1, 1)$) so that $x_k \succsim x_l$ iff $u(x_k) \geq u(x_l)$,
i.e. u represents the preference relation over $x_1, \dots, x_{n-1}$.
 - ▶ Now we define $u(x_n)$ (where $n \geq 2$).
 - ▶ If x_n is indifferent to x_k for some $k < n$, then assign $u(x_n) = u(x_k)$.
If not, then choose $u(x_n)$ to be $>$ than the maximum value of the set $\{u(x_k) | x_n \succ x_k\} \cup \{-1\}$ and $<$ than the minimum value of the set $\{u(x_k) | x_k \succ x_n\} \cup \{1\}$. This is possible since by transitivity all numbers in the first set are strictly below all numbers in the second set.
 - ▶ Thus, for any $k < n$ we have $x_n \succsim x_k$ iff $u(x_n) \geq u(x_k)$ and the function u extended to $\{x_1, \dots, x_n\}$ represents the preferences over those elements.

- ▶ To complete the proof that u represents $\succsim$, take any $x, y \in X$. For some k and l we have $x = x_k$ and $y = x_l$. Considering $n = \max\{k, l\}$ in the above implies that $x_k \succsim x_l$ iff $u(x_k) \geq u(x_l)$. □

Utility representation for $\succsim$ over uncountable sets

- ▶ often the set of alternatives is uncountable, e.g. a subset of the n -dimensional real space
- ▶ it might get crowded! mapping the n -dimensional real space onto the one dimensional real line in an order preserving way might be impossible depending on the preference relation, e.g. “too many” indifference sets
- ▶ not every rational preference relation over an uncountable set can be represented by a utility function
- ▶ first, an example of a simple and commonly used preference procedure on an uncountable set which does not have a utility representation
- ▶ second, identify a “natural” condition for a rational preference relation which guarantees the existence of a utility representation

Lexicographic Preferences

- ▶ "lexicographic preferences" order alternatives like a dictionary:
 - ▶ alternatives are ranked by comparing them according to a fixed sequence of different criteria until a criterion is reached by which the two alternatives can be distinguished and ranked;
 - ▶ dictionary analogy: for determining a word's position in the dictionary, the letters of the word are considered in sequence
- ▶ Formally,
 - ▶ let $(\succsim_k)_{k=1,\dots,K}$ be a K -tuple of rational preference relations over the set X ($\succsim_k$ is the rational preference relation over X according to the k th-criterion).
 - ▶ the lexicographic preferences $\succsim_L$ induced by those preferences are defined by:
 $x \succsim_L y$ iff (1) there is k^* such that for all $k < k^*$ we have $x \sim_k y$ and $x \succ_{k^*} y$ or (2) $x \sim_k y$ for all k .
(Verify that $\succsim_L$ is rational!)

Lexicographic Preferences: Example

- ▶ Let X be the unit square, that is, $X = [0, 1] \times [0, 1]$.
- ▶ Let $x \succsim_k y$ iff $x_k \geq y_k$.
- ▶ The lexicographic preferences $\succsim_L$ induced from $\succsim_1$ and $\succsim_2$ are:
 $(x_1, x_2) \succsim_L (y_1, y_2)$ iff $x_1 > y_1$ or both $x_1 = y_1$ and $x_2 \geq y_2$.
- ▶ These lexicographic preferences do not have a utility representation.
- ▶ Intuition: no two distinct alternatives are indifferent; each indifference set is a singleton; two dimensions of distinct indifference sets; however each indifference set has to be assigned (in an order-preserving way) a different utility number from the one-dimensional real line; impossibility!
- ▶ (It is not generally true that lexicographic preferences do not have a utility representation. This depends on X .)

Proof.

- ▶ Assume by contradiction that the function $u : X \rightarrow \mathbb{R}$ represents $\succsim_L$.
- ▶ For any $x_1 \in [0, 1]$, $(x_1, 1) \succ_L (x_1, 0)$ and thus $u(x_1, 1) > u(x_1, 0)$. Then we can pick a rational number $q(x_1)$ such that $u(x_1, 1) > q(x_1) > u(x_1, 0)$.
- ▶ Note that if $x'_1 > x_1$, then $(x'_1, 0) \succ_L (x_1, 1)$ and therefore $u(x'_1, 0) > u(x_1, 1)$.
- ▶ Thus $q(x'_1) > u(x'_1, 0) > u(x_1, 1) > q(x_1)$. Hence for any $x'_1 \neq x_1$, $q(x'_1) \neq q(x_1)$.
- ▶ Reminder: The function $f : X \rightarrow Z$ is a *one-to-one function* if $f(x) = f(y)$ implies that $x = y$.
- ▶ Then the function q provides a one-to-one function from the continuum $[0, 1]$ (uncountable) into the set of rational numbers (countable).
- ▶ Mathematical impossibility/contradiction: cardinality of the continuum is greater than that of the rational numbers.

Continuity of a rational $\succsim$ ensures a utility representation

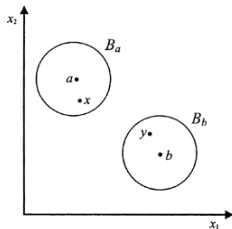
- ▶ *continuity* of a rational preference relation ensures the existence of a utility representation if X is an infinite subset of a Euclidean space (e.g. $X = \mathbb{R}^2$)
- ▶ assume in the following that X is an infinite subset of a Euclidean space
- ▶ basic intuition of the continuity property:
if a is preferred to b , then “small” deviations from a or from b will not change the ordering
- ▶ first, we will see three equivalent definitions of the continuity property of a rational preference relation
 - ▶ definition C1 reveals the intuition of the continuity property
 - ▶ definition C2 is most commonly used in economics
 - ▶ definition C3 introduces the concepts of upper and lower contour sets

Continuity of $\succsim$: Definition C1

- ▶ Intuition: if $a \in X$ is strictly preferred over $b \in X$, then small deviations from a are also strictly preferred over (small deviations from) b ; small deviations do not change ordering
- ▶ a ball around $a \in X$ with radius $r > 0$, denoted as $Ball(a, r)$, is the set of all points in X that are distanced less than r from a

Definition 3 (C1)

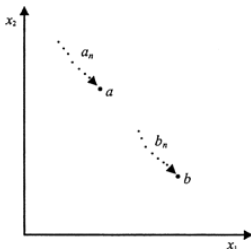
A rational preference relation $\succsim$ on X is continuous if whenever $a \succ b$ (namely, it is not true that $b \succsim a$), there are balls B_a and B_b around a and b , respectively, such that for all $x \in B_a$ and $y \in B_b$, $x \succ y$.



Continuity of $\succsim$: Definition C2

Definition 4 (C2)

A rational preference relation $\succsim$ on X is continuous if it is preserved under limits. That is, for any sequence of pairs $\{(a_n, b_n)\}_{n=1}^{\infty}$ with $a_n \succsim b_n$ for all n , $a_n \rightarrow a$ and $b_n \rightarrow b$, we have $a \succsim b$.



- C2 says the preferences cannot exhibit "jumps", e.g. if you prefer each element in $\{a_n\}_{n=1}^{\infty}$ to the corresponding element in $\{b_n\}_{n=1}^{\infty}$ (because of the convergence, the sequence gets arbitrarily close to a and b), you cannot reverse the preferences in the limit.

Equivalence of C1 and C2

Proposition 6

A rational preference relation on X satisfies C1 if and only if it satisfies C2.

Proof: C1 $\Rightarrow$ C2.

- ▶ Assume that $\succsim$ on X is continuous according to C1.
- ▶ Let $\{(a_n, b_n)\}_{n=1}^{\infty}$ be a sequence of pairs of elements in X with $a_n \succsim b_n$ for all n , $a_n \rightarrow a$ and $b_n \rightarrow b$.
- ▶ Suppose it is not true that $a \succsim b$ and thus $b \succ a$.
- ▶ Then, by C1, there exist two balls B_a and B_b around a and b , respectively, such that for all $y \in B_b$ and $x \in B_a$, $y \succ x$.
- ▶ Since $a_n \rightarrow a$ and $b_n \rightarrow b$, there is an integer N large enough such that for all $n > N$, both $b_n \in B_b$ and $a_n \in B_a$.
- ▶ Therefore, for all $n > N$, we have $b_n \succ a_n$, a contradiction.



Proof: $C2 \Rightarrow C1$.

- ▶ Assume that $\succsim$ is continuous according to C2.
- ▶ Let $a \succ b$.
- ▶ Assume by contradiction that C1 is not satisfied, that is for any balls B_a and B_b , there exist $x \in B_a$ and $y \in B_b$ such that $y \succsim x$.
- ▶ Then, for all n , there exist $a_n \in \text{Ball}(a, 1/n)$ and $b_n \in \text{Ball}(b, 1/n)$ such that $b_n \succsim a_n$.
- ▶ The sequence $\{(b_n, a_n)\}_{n=1}^{\infty}$ converges to (b, a) .
- ▶ Then, by C2, $b \succsim a$, a contradiction and C1 must be satisfied.



Continuity of $\succsim$: Definition C3

Definition 5 (C3)

A rational preference relation $\succsim$ is continuous if, for all $x \in X$, the upper contour set $\{y \in X : y \succsim x\}$ and the lower contour set $\{y \in X : x \succsim y\}$ are closed.

- ▶ reminder: a set is closed if it includes its boundaries
- ▶ Given a preference relation $\succsim$, the upper contour set of $x \in X$ contains all $y \in X$ which are weakly preferred to x , and the lower contour set of x contains all $y \in X$ to which x is weakly preferred.
- ▶ Given a rational preference relation, the union of the lower and upper contour set of any x is equal to X (because of completeness and reflexivity), and the intersection is the indifference set with x .

Equivalence of C1 and C3

Proposition 7

A rational preference relation $\succsim$ on X satisfies C3 if and only if it satisfies C1.

Proof.

- ▶ First note that for all $x \in X$,
 $X \setminus \{y \in X : x \succsim y\} = \{y \in X : y \succ x\}$ and
 $X \setminus \{y \in X : y \succsim x\} = \{y \in X : x \succ y\}$
(complements to $\{y \in X : x \succsim y\}$ and $\{y \in X : y \succsim x\}$,
respectively)
by completeness and reflexivity of $\succsim$.
- ▶ $\{y \in X : y \succ x\}$ and $\{y \in X : x \succ y\}$ are open sets if and only if $\{y \in X : x \succsim y\}$ and $\{y \in X : y \succsim x\}$ are closed sets.

Proof cont'd

- ▶ Show:
 $\succsim$ satisfies C1 $\Rightarrow \{y \in X : y \succ x\}$ and $\{y \in X : x \succ y\}$ are open sets for all $x \in X$ ($\Rightarrow \succsim$ satisfies C3):
 - ▶ Let $a \in \{y \in X : y \succ x\}$.
 - ▶ By C1, there exists $Ball(a, r)$ such that $Ball(a, r) \subseteq \{y \in X : y \succ x\}$.
 - ▶ This is the definition for $\{y \in X : y \succ x\}$ to be an open set.
 - ▶ Equivalently, for $\{y \in X : x \succ y\}$.

Proof cont'd

- ▶ ($\succsim$ satisfies C3 $\Rightarrow$) $\{y \in X : y \succ x\}$ and $\{y \in X : x \succ y\}$ are open sets for all $x \in X \Rightarrow \succsim$ satisfies C1:
 - ▶ We use the notation $B \succsim x$ to mean that $y \succsim x$ for all $y \in B$.
 - ▶ Consider $a, b \in X$ such that $a \succ b$.
 - ▶ Assume there exists $z \in X$ such that $a \succ z \succ b$.
 - ▶ Then $a \in \{x \in X : x \succ z\}$ and $b \in \{x \in X : z \succ x\}$.
 - ▶ $\{x \in X : x \succ z\}$ and $\{x \in X : z \succ x\}$ are open sets.
 - ▶ Thus there exists $\epsilon > 0$ such that $Ball(a, \epsilon) \subset \{x \in X : x \succ z\}$ and $Ball(b, \epsilon) \subset \{x \in X : z \succ x\}$, that is $Ball(a, \epsilon) \succ z$ and $z \succ Ball(b, \epsilon)$.
 - ▶ By transitivity, $x \succ y$ for all $x \in Ball(a, \epsilon)$ and all $y \in Ball(b, \epsilon)$.

- ▶ Assume there does not exist $z \in X$ such that $a \succ z \succ b$.
- ▶ By assumption $b \in \{x \in X : a \succ x\}$ and $a \in \{x \in X : x \succ b\}$.
- ▶ $\{x \in X : a \succ x\}$ and $\{x \in X : x \succ b\}$ are open sets.
- ▶ Thus there exists $\epsilon > 0$ such that $Ball(b, \epsilon) \subset \{x \in X : a \succ x\}$ and $Ball(a, \epsilon) \subset \{x \in X : x \succ b\}$, that is $a \succ Ball(b, \epsilon)$ and $Ball(a, \epsilon) \succ b$.
- ▶ Since there is no $z \in X$ (and thus not in $Ball(b, \epsilon)$ and not in $Ball(a, \epsilon)$) such that $a \succ z \succ b$, it must be that $Ball(a, \epsilon) \precsim a$ and $b \precsim Ball(b, \epsilon)$.
- ▶ Thus $Ball(a, \epsilon) \precsim a \succ b \precsim Ball(b, \epsilon)$.
- ▶ And by transitivity, $x \succ y$ for all $x \in Ball(a, \epsilon)$ and all $y \in Ball(b, \epsilon)$.



Our example of lexicographic preferences was not continuous!

- ▶ Remember the example:
Let $X = [0, 1] \times [0, 1]$ and $\succsim_L$ be defined as $(x_1, x_2) \succsim_L (y_1, y_2)$ iff $x_1 > y_1$ or both $x_1 = y_1$ and $x_2 \geq y_2$.
- ▶ Then $a = (0, 1) \succ (0, 0) = b$ but there are no B_a and B_b such that $x \succ y$ for all $x \in B_a$ and $y \in B_b$ because e.g. $(y_1, 0) \succ a$ for $y_1 > 0$.
- ▶ Thus C1 does not hold and $\succsim_L$ is not continuous.